QUASI-COHERENT SHEAVES ON AN AFFINE SCHEME COME FROM THEIR GLOBAL SECTIONS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1504-1604, following the statement at 1495-1502.

Editorial note. The sentence 'We leave it to the reader to show that m -> s is a R-module map' (~1595-1596) is a mechanical linearity check after the map is displayed explicitly; flagged in the notes.

Editorial note. The triple-overlap cocycle 'It is clear' (~1534-1535) is read off the constructive definition of psi_ij as a composition of restrictions of isomorphisms.

Allowed standard prerequisites (editorial).

- Local representation of quasi-coherent modules. Modules, Lemma modules-lemma-quasi-coherent-module (cross-chapter, cited at ~1510). Gives a qc-neighborhood cover on which F is a tilde sheaf.
- Standard open refinement. lemma-standard-open (same chapter, cited at ~1516). Refines the neighborhood cover to a standard open cover D(f_i).
- Comparison of the two tilde constructions. lemma-compare-constructions (same chapter, cited at ~1548, ~1551, ~1597). Identifies tilde sheaves with their modules on affines and converts sheaf maps to module maps.
- Glueing of modules. Algebra, Lemma algebra-lemma-glue-modules (cross-chapter, cited at ~1572). Produces M with M_i = M_{f_i} compatible with the psi_ij.
- Glueing sheaves from data. Sheaves, Section sheaves-section-glueing-sheaves, in particular sheaves-lemma-mapping-property-glue (cross-chapter, cited at ~1580-1582). Frames F and tilde(M) as glueing data over the covering.

Source reference. Original reference 00EQ: focused-seeds/geometry/S9-schemes-quasi-coherent-affine/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01I7: sources/stacks-schemes/pinned-original-source.tex, source lines 1295–1295. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01HS: sources/stacks-schemes/pinned-original-source.tex, source lines 737–737. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

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to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01BK: focused-seeds/geometry/S9-schemes-quasi-coherent-affine/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00AN: focused-seeds/geometry/S9-schemes-quasi-coherent-affine/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00AK: focused-seeds/geometry/S9-schemes-quasi-coherent-affine/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0068: focused-seeds/geometry/S9-schemes-quasi-coherent-affine/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $(X, \mathcal{O}_X) = (\mathrm{Spec}(R), \mathcal{O}_{\mathrm{Spec}(R)})$ be an affine scheme. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Then $\mathcal{F}$ is isomorphic to the sheaf associated to the R-module $\Gamma(X, \mathcal{F})$.*

Proof. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Since every standard open $D(f)$ is quasi-compact we see that X is locally quasi-compact, i.e., every point has a fundamental system of quasi-compact neighbourhoods, see Topology, Definition 0068. Hence by Modules, Lemma 01BK for every prime $\mathfrak{p} \subset R$ corresponding to $x \in X$ there exists an open neighbourhood $x \in U \subset X$ such that $\mathcal{F}|_U$ is isomorphic to the quasi-coherent sheaf associated to some $\mathcal{O}_X(U)$-module M. In other words, we get an open covering by U's with this property. By Lemma 01HS for example we can refine this covering to a standard open covering. Thus we get a covering $\mathrm{Spec}(R) = \bigcup D(f_i)$ and R_{f_i}-modules M_i and isomorphisms $\varphi_i : \mathcal{F}|_{D(f_i)} \to \mathcal{F}_{M_i}$ for some R_{f_i}-module M_i. On the overlaps we get isomorphisms

$$\mathcal{F}_{M_i}|_{D(f_i f_j)} \xrightarrow{\varphi_i^{-1}|_{D(f_i f_j)}} \mathcal{F}|_{D(f_i f_j)} \xrightarrow{\varphi_j|_{D(f_i f_j)}} \mathcal{F}_{M_j}|_{D(f_i f_j)}.$$

Let us denote these ψ_{ij}. It is clear that we have the cocycle condition

$$\psi_{jk}|_{D(f_i f_j f_k)} \circ \psi_{ij}|_{D(f_i f_j f_k)} = \psi_{ik}|_{D(f_i f_j f_k)}$$

on triple overlaps.

Recall that each of the open subspaces $D(f_i)$, $D(f_i f_j)$, $D(f_i f_j f_k)$ is an affine scheme. Hence the sheaves $\mathcal{F}_{M_i}$ are isomorphic to the sheaves $\widetilde{M}_i$ by Lemma 01I7

above. In particular we see that $\mathcal{F}_{M_i}(D(f_if_j)) = (M_i)_{f_j}$, etc. Also by Lemma 01I7 above we see that ψ_{ij} corresponds to a unique $R_{f_if_j}$-module isomorphism

$$\psi_{ij} : (M_i)_{f_j} \longrightarrow (M_j)_{f_i}$$

namely, the effect of ψ_{ij} on sections over $D(f_if_j)$. Moreover these then satisfy the cocycle condition that

$$\begin{array}{ccc} (M_i)_{f_jf_k} & \xrightarrow{\psi_{ik}} & (M_k)_{f_if_j} \\ & {\scriptstyle\psi_{ij}} \searrow \quad \nearrow {\scriptstyle\psi_{jk}} & \\ & (M_j)_{f_if_k} & \end{array}$$

commutes (for any triple i, j, k).

Now Algebra, Lemma 00EQ shows that there exist an R-module M such that $M_i = M_{f_i}$ compatible with the morphisms ψ_{ij}. Consider $\mathcal{F}_M = \widetilde{M}$. At this point it is a formality to show that $\widetilde{M}$ is isomorphic to the quasi-coherent sheaf $\mathcal{F}$ we started out with. Namely, the sheaves $\mathcal{F}$ and $\widetilde{M}$ give rise to isomorphic sets of glueing data of sheaves of $\mathcal{O}_X$-modules with respect to the covering $X = \bigcup D(f_i)$, see Sheaves, Section 00AK and in particular Lemma 00AN. Explicitly, in the current situation, this boils down to the following argument: Let us construct an R-module map

$$M \longrightarrow \Gamma(X, \mathcal{F}).$$

Namely, given $m \in M$ we get $m_i = m/1 \in M_{f_i} = M_i$ by construction of M. By construction of M_i this corresponds to a section $s_i \in \mathcal{F}(U_i)$. (Namely, $\varphi_i^{-1}(m_i)$.) We claim that $s_i|_{D(f_if_j)} = s_j|_{D(f_if_j)}$. This is true because, by construction of M, we have $\psi_{ij}(m_i) = m_j$, and by the construction of the ψ_{ij}. By the sheaf condition of $\mathcal{F}$ this collection of sections gives rise to a unique section s of $\mathcal{F}$ over X. We leave it to the reader to show that $m \mapsto s$ is a R-module map. By Lemma 01I7 we obtain an associated $\mathcal{O}_X$-module map

$$\widetilde{M} \longrightarrow \mathcal{F}.$$

By construction this map reduces to the isomorphisms φ_i^{-1} on each $D(f_i)$ and hence is an isomorphism. □